Sugar-Dark Side of Sweetness

(Diabetes)

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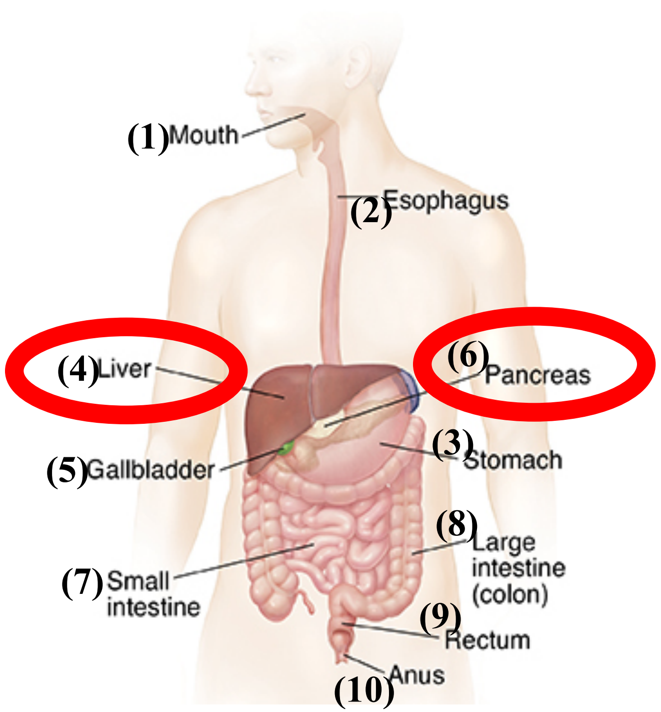
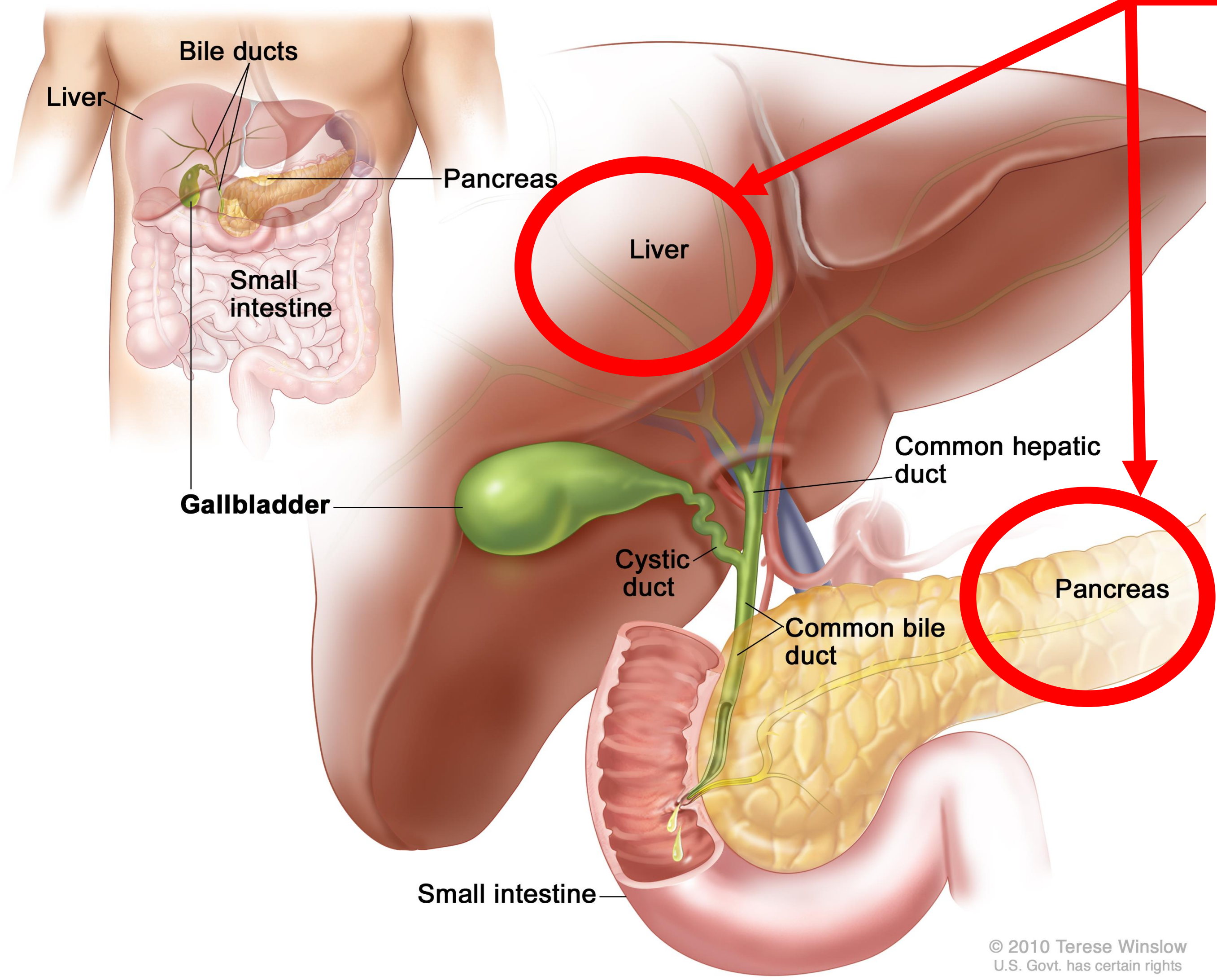
Diabetes = Large excess Blood glucose

How Blood Glucose is Regulated in our Body?

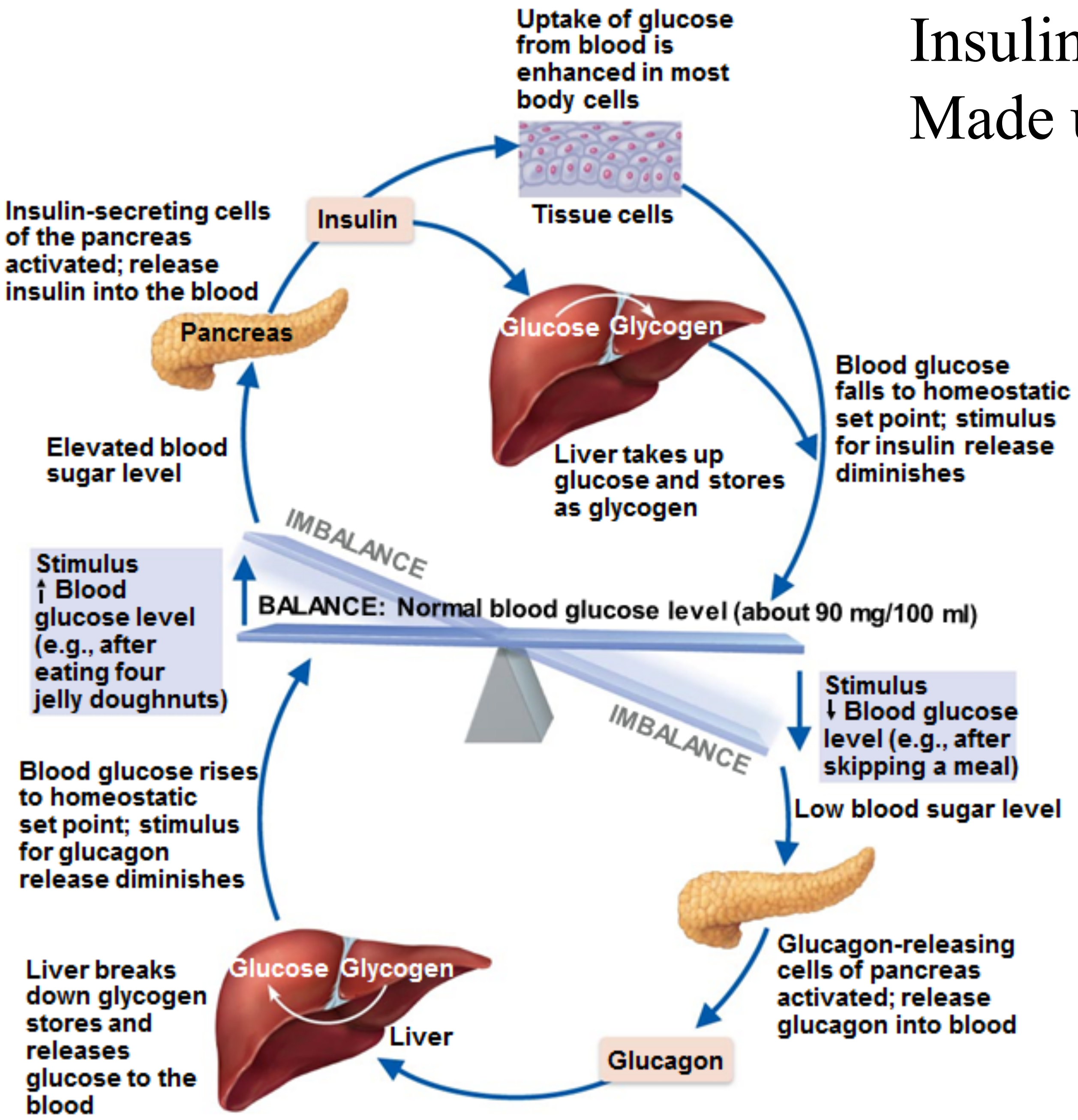
Story of



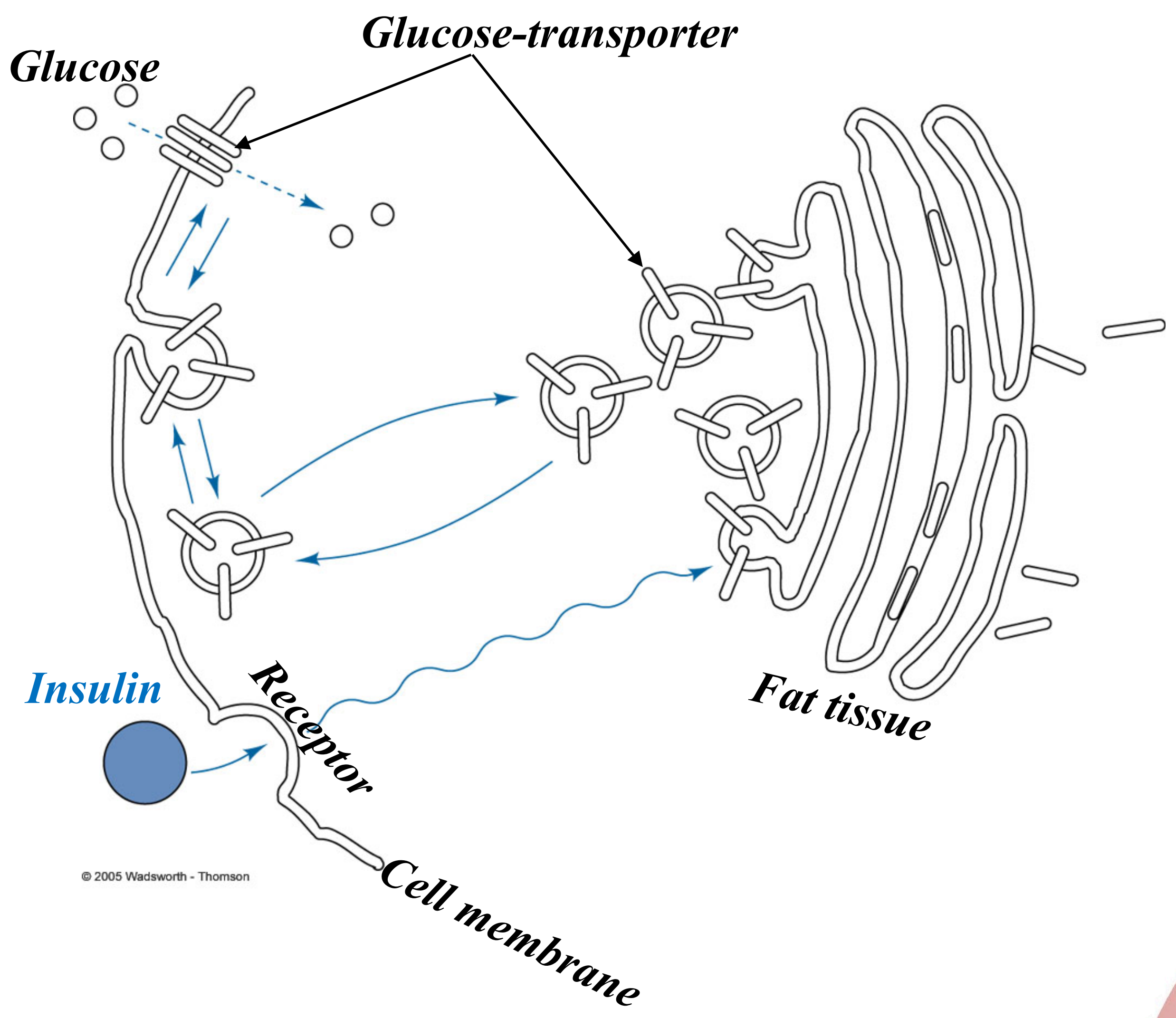
Digestive System (lecture 3)



Insulin and glucagon are are hormones: Made up of amino-acids (“small protein”)

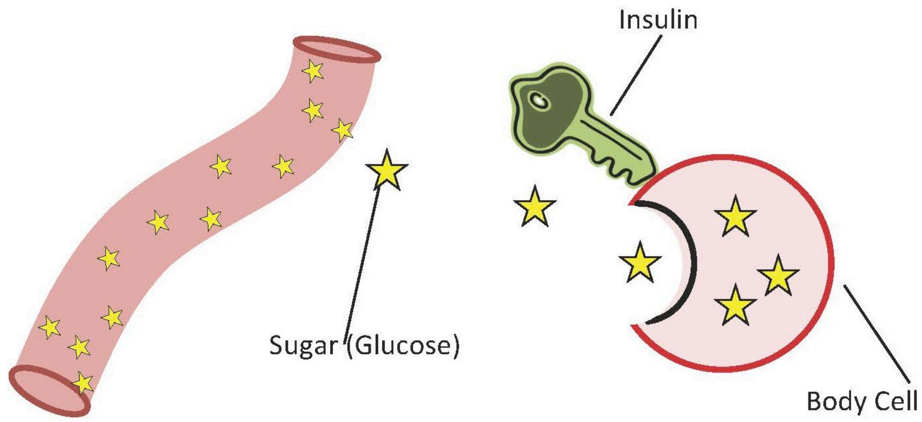


Hormones are signalling molecules that regulate (up or Down) the activity of cells.



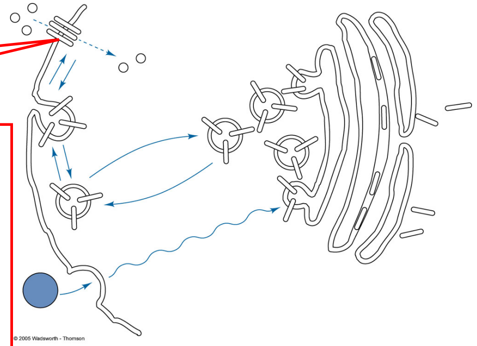
Proposed mode of insulin action on glucose transport:

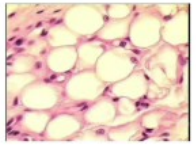
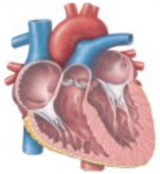
(1) The binding of insulin to its specific receptor induces a signal of **as yet unknown nature** that stimulates the translocation of vesicles containing the glucose transporters to the cell membrane.
(2) The vesicles fuse with the cell membrane, releasing the transporters and allowing them to position themselves in the membrane.
(3) On removal of insulin from its receptor, membrane-bound transporters are retranslocated back to the intracellular pool by an endocytosis-like mechanism.



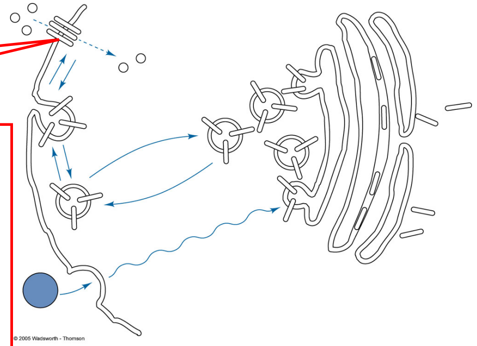
When a cell needs energy, insulin acts as a key to unlock the cell. This opens the cell so sugar can enter and be used for energy.

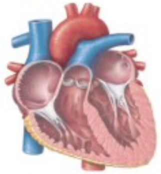
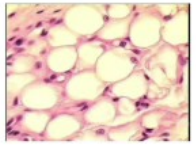
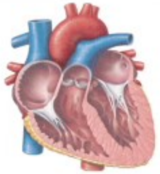
Few Glucose Transporters



GLUT4	• Skeletal Muscle • Adipose Tissue • Heart   	**INSULIN DEPENDENT**

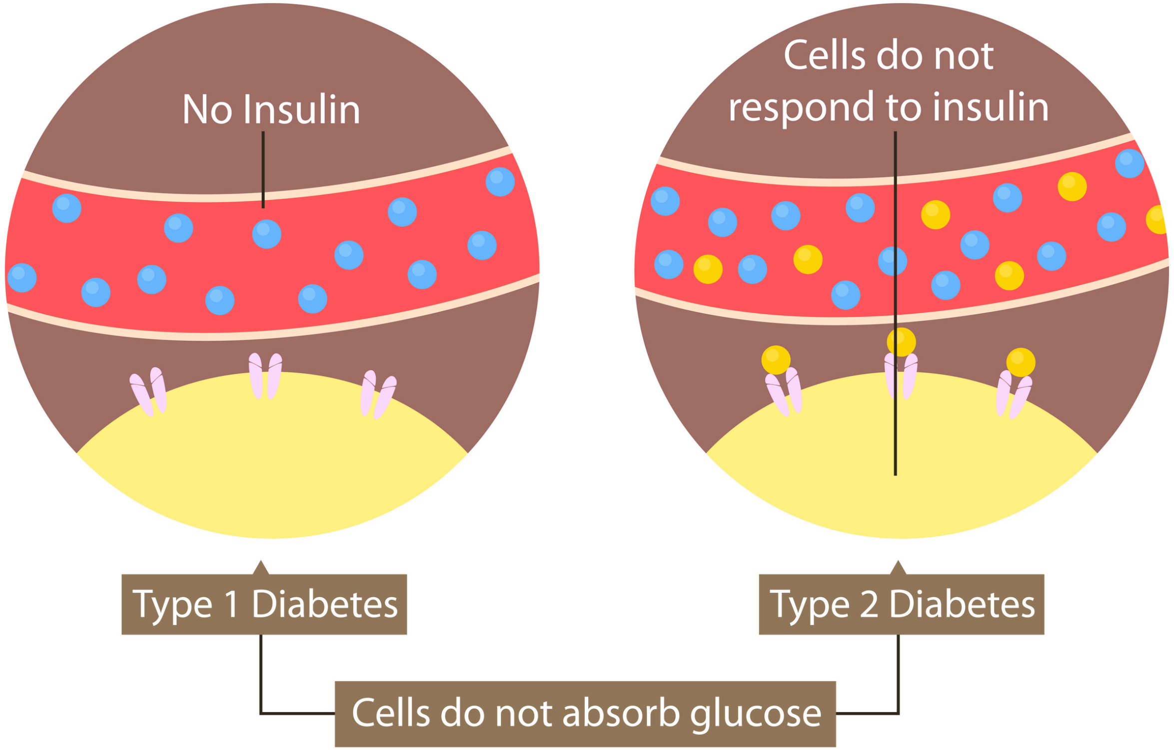
Few Glucose Transporters



GLUT1	• Blood • Blood-Brain Barrier • Heart (lesser extent)  	• Insulin-Independent
GLUT2	• Liver • Pancreas • Small Intestine  	• Insulin-Independent • High K_m • Low Affinity
GLUT3	• Brain • Neurons • Sperm 	• Insulin-Independent • Low K_m • High Affinity
GLUT4	• Skeletal Muscle • Adipose Tissue • Heart   	**INSULIN DEPENDENT**

Diabetes = Large excess Blood glucose

Types of Diabetes

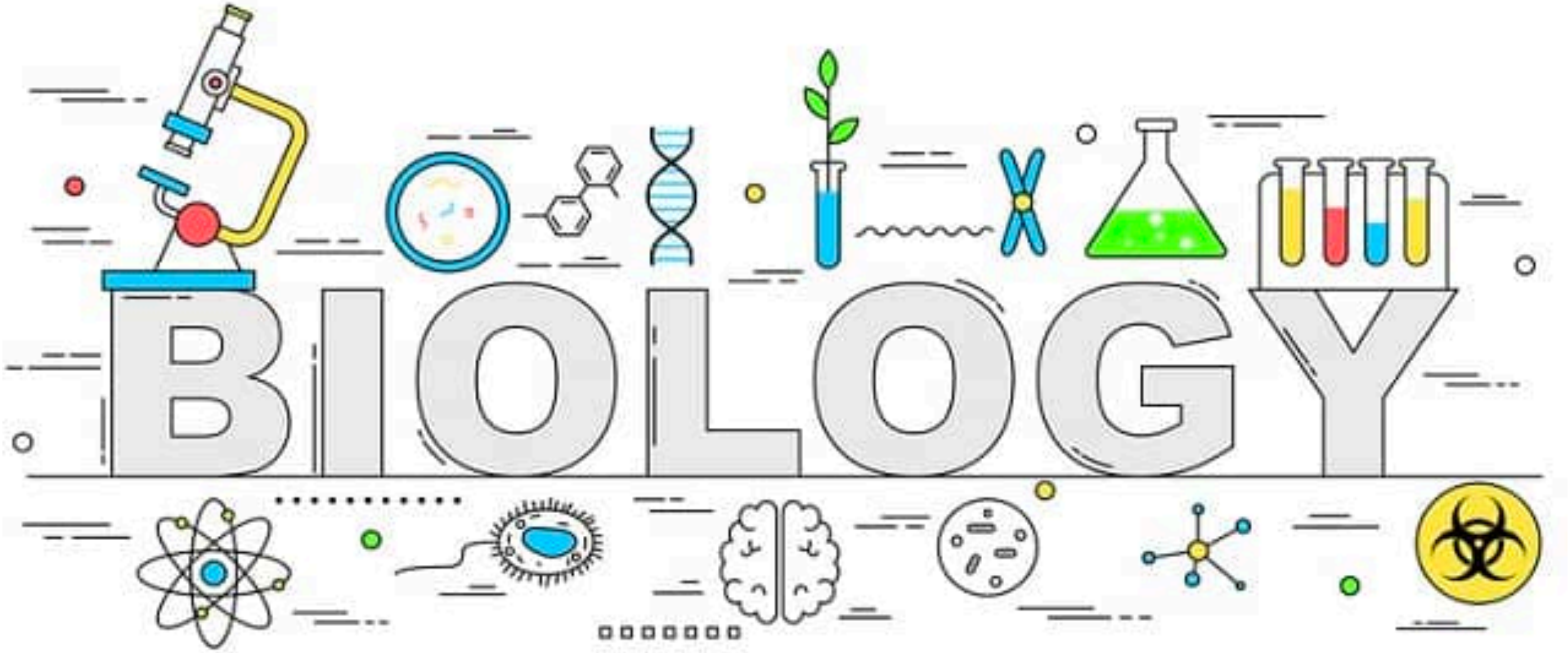


Take home

1. Blood glucose regulation in our body: STORY of Liver & Pancreas

2. Diabetes = Large excess Blood glucose

3. Cause of Diabetes:
   - NO insulin production in the body (Type 1)
   - Insulin is produced but cells can-not absorb glucose (Type 2)



Next: ***Sugar-Dark Side of Sweetness (Cont…)***